

RONISH PRAJAPATI

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PROFESSIONAL SUMMARY

Passionate AI/ML student with hands-on internship and project experience in end-to-end ML pipelines, Deep Learning architectures and Agentic RAG systems. Proven ability to build APIs with FastAPI, optimize inference pipelines on local hardware, and deploy computer vision and NLP models. Driven to solve complex problems, learn from experienced mentors and collaborate with like-minded individuals on AI-based solutions.

EDUCATION

London Metropolitan University / Islington College	B.Sc (Hons) Computing with AI	2023 – Present
Campion Academy	Higher Secondary / A-Levels	2020 – 2022

EXPERIENCE

Tanvi Tech Pvt. Ltd.	Machine Learning Intern	Sep 2025 – Dec 2025
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- **Built an end-to-end Devanagari OCR pipeline** comparing a Custom CNN model, a pre-trained vision network, and a fine-tuned architecture on the Devanagari dataset to classify characters.
- **Developed the devanagari_ocr_app via FastAPI** to wrap the models, exposing highly performant API endpoints for seamless image character classification.
- **Executed intensive performance benchmarking** and streamlined image data preprocessing workflows to ensure robust evaluation and deployment readiness.

PROJECTS

RAG Engine | Python, FastAPI, Qdrant, Redis, SQLite, Ollama

- **Developed an Agentic RAG service** natively using local open-source models (Llama 3.1 & Nomic Embedding), entirely bypassing RAG frameworks like LangChain or LlamaIndex.
- **Implemented a dual-engine data persistence** strategy incorporating a high-speed Redis cluster for multi-turn session tracking and SQLite for reliable transactional metadata storage.
- **Optimized inference pipeline** cold-start latency by designing a **FastAPI lifespan** event hook that automatically preloads and pins local architectures into GPU VRAM on system boot.

Hansei | Python, PyTorch, Node.js, FastAPI, Qdrant, React Native

- **Architected and deployed a cross-platform 'clinical companion' app** featuring a React Native mobile frontend, Node.js backend, and a high-performance FastAPI AI engine.
- **Integrated a Retrieval-Augmented Generation (RAG) pipeline** backed by a Qdrant vector database to cross-reference user queries against structured CBT guidelines for reliable chat generation.
- **Designed a Transformer-based deep learning module** utilizing the GoEmotions dataset to support real-time text emotion tracking and automated cognitive reframing.

Deforestation Detection | Python, PyTorch, ResNet-50, OpenCV, Gradio

- **Developed an automated land cover classification system** utilizing the EuroSAT dataset to track and monitor deforestation patterns from satellite imagery.
- **Achieved a ~98% test accuracy** with the fine-tuned ResNet-50 model, successfully resolving complex feature classification between forest and pasture terrains.

GoEmotion Classification | Python, PyTorch, Transformers, Hugging Face, Sci-Kit Learn

- **Implemented and benchmarked three deep learning architectures**—Bi-LSTM, Seq2Seq, and an Encoder-only Transformer—to optimize multi-label performance.
- **Achieved a 0.5098 Macro F1-Score** with the Transformer model, successfully outperforming the established Google baseline.

TECHNICAL SKILLS

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- **AI & Agentic Frameworks:** RAG-based AI Engines, HuggingFace, Transformers, LLMs, Computer Vision (CNNs, ResNet, OCR), NLP, Qdrant
 - **ML & Data Science:** PyTorch, TensorFlow, Sci-Kit Learn, Neural Network Architectures from Scratch, Data Preprocessing
 - **Programming & Backend:** Python (Asynchronous), JavaScript (Node.js), FastAPI, Java, Data Structures & Algorithms
 - **Infrastructure & Tools:** Docker, Linux, Windows, Secure API Integration, Git, VS Code, Jupyter, Colab

CERTIFICATIONS

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- **Certifications:** [Supervised Learning](#), [Advanced Learning Algorithms](#), [Unsupervised Learning](#) (DeepLearning.AI / Stanford).